

Chapter 7 Test model

- **70 Multiple Choice Questions (MCQs)**
 - **30 True/False Questions**
 - **Difficulty:** 70% Hard, 30% Very Hard
 - **Format:** Clear numbering, line separator (___) between questions
 - **Answers** provided **after all questions**
-

Multiple Choice Questions (1–70)

1. In SQL's three-valued logic, what is the result of the expression `NULL = NULL` ?
 - a. TRUE
 - b. FALSE
 - c. UNKNOWN
 - d. ERROR

2. Which SQL clause is used to check for the presence of a NULL value in a column?
 - a. EQUALS NULL
 - b. IS NULL
 - c. NULL CHECK
 - d. EXISTS NULL

3. In a correlated nested query, the inner query:
 - a. Executes once before the outer query
 - b. Is independent of the outer query
 - c. Is evaluated once for each tuple of the outer query
 - d. Cannot reference attributes from the outer query

4. Which of the following is **not** a valid interpretation of NULL in SQL?
 - a. Unknown value
 - b. Zero value
 - c. Unavailable value
 - d. Not applicable

5. The operator `= ANY` is equivalent to which of the following?

- a. NOT IN
 - b. EXISTS
 - c. IN
 - d. ALL
-

6. Consider the query:

```
SELECT * FROM EMPLOYEE WHERE Salary > ALL (SELECT Salary FROM EMPLOYEE WHERE Dno = 5);
```

This returns employees whose salary is:

- a. Greater than the highest salary in department 5
 - b. Greater than the average salary in department 5
 - c. Greater than any salary in department 5
 - d. Not in department 5 and higher paid than everyone in dept 5
-

7. Which of the following correctly describes a **correlated subquery**?

- a. It can be evaluated independently of the outer query
 - b. It uses aggregate functions only
 - c. It references attributes from the outer query
 - d. It must appear in the SELECT clause
-

8. In SQL, comparing any value with NULL using standard comparison operators (e.g., =, <) yields:

- a. TRUE
 - b. FALSE
 - c. UNKNOWN
 - d. NULL
-

9. Which keyword is used to ensure a value is **not equal to any** value returned by a subquery?

- a. `<> ALL`
- b. NOT IN
- c. `<> ANY`
- d. `!= SOME`

-
10. The expression `X > ALL (subquery)` is TRUE only if:
- a. X is greater than at least one value in the subquery result
 - b. X is greater than the average of the subquery result
 - c. X is greater than every value in the subquery result
 - d. The subquery returns no rows
-

11. What happens if a subquery used with `NOT IN` returns a NULL value?
- a. The entire condition evaluates to UNKNOWN
 - b. The query returns an error
 - c. NULLs are ignored
 - d. The condition becomes TRUE
-

12. Which of the following is **true** about tuple variables (aliases) in SQL?
- a. They are optional and rarely used
 - b. They are required only in nested queries
 - c. They help avoid ambiguity in queries with multiple table references
 - d. They cannot be used in the WHERE clause
-

13. In three-valued logic, what is the result of `TRUE AND UNKNOWN` ?
- a. TRUE
 - b. FALSE
 - c. UNKNOWN
 - d. NULL
-

14. Which operator can be used to compare a single value against a set of values returned by a subquery, returning TRUE if it matches **at least one**?
- a. `= ALL`
 - b. `= ANY`
 - c. `<> ALL`
 - d. `NOT EXISTS`
-

15. The query:

```
SELECT Fname FROM EMPLOYEE E WHERE EXISTS (SELECT * FROM DEPENDENT D WHERE D.Essn = E.Ssn);
```

is an example of:

- a. Uncorrelated subquery
 - b. Correlated subquery
 - c. Aggregate query
 - d. Outer join
-

16. Which of the following best describes the logical evaluation of `WHERE column IS NOT NULL` ?

- a. Filters out rows where column is UNKNOWN
 - b. Includes only rows where column has a definite non-NULL value
 - c. Treats NULL as zero
 - d. Converts NULL to empty string
-

17. In SQL, if a subquery in an `IN` predicate returns an empty set, the condition evaluates to:

- a. TRUE
 - b. FALSE
 - c. UNKNOWN
 - d. ERROR
-

18. Which of the following statements about `= SOME` is correct?

- a. It is syntactically invalid
 - b. It is equivalent to `= ANY`
 - c. It requires the subquery to return exactly one row
 - d. It is the opposite of `<> ALL`
-

19. To rewrite a correlated query as a join, which of the following must be true?

- a. The correlation condition must be expressible as a join predicate
 - b. The subquery must use aggregate functions
 - c. The outer query must have GROUP BY
 - d. The tables must have identical schemas
-

20. Which of the following is **not** a reason to use tuple aliases in SQL?

- a. To shorten table names
 - b. To distinguish between multiple instances of the same table
 - c. To avoid ambiguity in attribute references
 - d. To enable recursive queries
-

21. The truth table for **OR** in three-valued logic includes which of the following outcomes?

- a. UNKNOWN OR FALSE = UNKNOWN
 - b. UNKNOWN OR TRUE = FALSE
 - c. UNKNOWN OR UNKNOWN = FALSE
 - d. FALSE OR FALSE = UNKNOWN
-

22. In the query:

```
SELECT * FROM EMPLOYEE WHERE (Fname, Lname) IN (SELECT Dependent_name, Sex
FROM DEPENDENT);
```

What is the likely issue?

- a. Type mismatch between compared attributes
 - b. Syntax error: tuples cannot be used with IN
 - c. Subquery returns more than one row
 - d. No issue; valid SQL
-

23. Which of the following is a **correct** use of tuple comparison in SQL?

- a. (A, B) = (SELECT X, Y FROM T)
 - b. (A, B) IN (SELECT X FROM T)
 - c. (A) = (B, C)
 - d. A, B = SELECT X, Y FROM T
-

24. If a condition in the WHERE clause evaluates to UNKNOWN for a row, what happens to that row?

- a. It is included in the result
 - b. It is excluded from the result
 - c. It causes a runtime error
 - d. It is flagged for manual review
-

25. Which SQL construct **cannot** be used to handle NULL-related logic explicitly?

- a. IS NULL
 - b. COALESCE
 - c. = NULL
 - d. CASE
-

26. The expression `NOT (X = ANY (subquery))` is logically equivalent to:

- a. `X <> ALL (subquery)`
 - b. `X = ALL (subquery)`
 - c. `NOT EXISTS (subquery)`
 - d. `X NOT IN (subquery)` — **only if no NULLs**
-

27. Why is `X NOT IN (subquery)` problematic when the subquery may return NULLs?

- a. It throws a syntax error
 - b. It always returns TRUE
 - c. It evaluates to UNKNOWN for all X, returning no rows
 - d. It ignores NULLs automatically
-

28. Which of the following is **true** about the evaluation of nested queries with `ALL` ?

- a. If the subquery returns an empty set, `> ALL (...)` is FALSE
 - b. If the subquery returns an empty set, `> ALL (...)` is TRUE
 - c. `ALL` requires at least two values
 - d. `ALL` cannot be used with string comparisons
-

29. In three-valued logic, what is `NOT UNKNOWN` ?

- a. TRUE
 - b. FALSE
 - c. UNKNOWN
 - d. NULL
-

30. Which of the following queries is **invalid** due to improper tuple usage?

- a. `WHERE (Ssn, Dno) = (SELECT Mgr_ssn, Dnumber FROM DEPARTMENT WHERE Dname = 'Research')`
- b. `WHERE (Salary) > (SELECT MAX(Salary) FROM EMPLOYEE)`

- c. WHERE (Fname, Lname) IN (SELECT Fname, Lname FROM EMPLOYEE)
 - d. WHERE (Ssn) = (SELECT Essn, Dependent_name FROM DEPENDENT)
-

31. The query:

```
SELECT E.Fname FROM EMPLOYEE E WHERE E.Ssn IN (SELECT D.Essn FROM DEPENDENT
D WHERE D.Sex = E.Sex);
```

is:

- a. Valid and correlated
 - b. Invalid due to correlation in subquery without EXISTS
 - c. Valid but uncorrelated
 - d. Invalid because E.Sex is not in SELECT of subquery
-

32. Which of the following best describes the semantics of EXISTS (subquery) ?

- a. Returns TRUE if subquery returns at least one row
 - b. Returns TRUE only if subquery returns exactly one row
 - c. Returns the number of rows in subquery
 - d. Equivalent to IN (*)
-

33. In SQL, what is the result of 5 > ALL (SELECT Salary FROM EMPLOYEE WHERE 1=0) ?

- a. TRUE
 - b. FALSE
 - c. UNKNOWN
 - d. ERROR
-

34. Which of the following is a **safe** alternative to NOT IN when NULLs may exist?

- a. <> ALL
 - b. NOT EXISTS
 - c. IS NOT NULL AND NOT IN
 - d. Both b and c
-

35. The use of aliases in a self-join is:

- a. Optional but recommended
- b. Mandatory to distinguish table instances

- c. Not allowed in standard SQL
 - d. Only needed in subqueries
-

36. Which logical operator in three-valued logic can **never** return TRUE if one operand is UNKNOWN?
- a. AND
 - b. OR
 - c. NOT
 - d. None of the above
-

37. Consider:

```
SELECT * FROM R WHERE A = (SELECT B FROM S);
```

If the subquery returns **more than one row**, the query:

- a. Returns all matching rows
 - b. Returns an error
 - c. Compares A to the first row only
 - d. Uses aggregate implicitly
-

38. Which of the following is **not** a characteristic of correlated subqueries?
- a. They reference outer query attributes
 - b. They are executed once per outer row
 - c. They can always be rewritten as joins
 - d. They often use EXISTS or IN
-

39. The condition `column <> NULL` in SQL:
- a. Filters out NULL values
 - b. Always evaluates to UNKNOWN
 - c. Is equivalent to `column IS NOT NULL`
 - d. Causes a syntax error
-

40. In the absence of NULLs, which pair of expressions are **equivalent**?
- a. `X NOT IN (Q)` and `X <> ALL (Q)`
 - b. `X IN (Q)` and `X = ALL (Q)`

- c. `X > ANY (Q)` and `X > ALL (Q)`
 - d. `EXISTS (Q)` and `IN (Q)`
-

41. Which of the following is **true** about SQL's treatment of duplicate NULLs?

- a. All NULLs are considered equal
 - b. Each NULL is distinct from every other NULL
 - c. NULLs are ignored in comparisons
 - d. NULLs are converted to default values
-

42. The expression `UNKNOWN OR TRUE` evaluates to:

- a. TRUE
 - b. FALSE
 - c. UNKNOWN
 - d. NULL
-

43. Which clause **cannot** contain a correlated subquery?

- a. SELECT
 - b. FROM
 - c. WHERE
 - d. HAVING
-

44. A subquery that returns a single value is called:

- a. Scalar subquery
 - b. Vector subquery
 - c. Set subquery
 - d. Tuple subquery
-

45. Which of the following queries correctly finds employees with no dependents?

- a. `SELECT * FROM EMPLOYEE WHERE Ssn NOT IN (SELECT Essn FROM DEPENDENT);`
 - b. `SELECT * FROM EMPLOYEE E WHERE NOT EXISTS (SELECT * FROM DEPENDENT D WHERE D.Essn = E.Ssn);`
 - c. `SELECT * FROM EMPLOYEE WHERE Ssn <> ALL (SELECT Essn FROM DEPENDENT);`
 - d. Both b and c — **but a fails if NULLs in Essn**
-

46. In three-valued logic, FALSE AND UNKNOWN evaluates to:

- a. FALSE
 - b. UNKNOWN
 - c. TRUE
 - d. NULL
-

47. Which of the following is **not** allowed in standard SQL?

- a. (A, B) = (C, D)
 - b. (A, B) IN (SELECT X, Y FROM T)
 - c. (A) = (SELECT X, Y FROM T)
 - d. A = (SELECT X FROM T WHERE condition)
-

48. The query:

```
SELECT * FROM EMPLOYEE WHERE Salary > (SELECT AVG(Salary) FROM EMPLOYEE);
```

uses a:

- a. Correlated scalar subquery
 - b. Uncorrelated scalar subquery
 - c. Correlated row subquery
 - d. Uncorrelated table subquery
-

49. If a subquery used with = ANY returns NULL and other values, and the comparison value is 10, what is 10 = ANY (... , NULL, ...)?

- a. TRUE if 10 matches any non-NULL value
 - b. UNKNOWN
 - c. FALSE
 - d. ERROR
-

50. Which of the following is **true** about the IN operator and NULLs?

- a. If the left side is NULL, NULL IN (1, 2) is TRUE
 - b. If the list contains NULL, it affects the result only if no match is found
 - c. IN ignores NULLs in the list
 - d. IN treats NULL as zero
-

51. The safest way to test for inequality against a subquery that might contain NULLs is:

- a. Use NOT IN
 - b. Use <> ALL
 - c. Use NOT EXISTS
 - d. Use IS NOT NULL AND NOT IN
-

52. Which of the following is a **valid** correlated query?

- a. SELECT * FROM EMP E WHERE E.Dno IN (SELECT Dnumber FROM DEPT);
 - b. SELECT * FROM EMP E WHERE EXISTS (SELECT * FROM DEPT D WHERE D.Mgr_ssn = E.Ssn);
 - c. SELECT * FROM EMP WHERE Ssn = (SELECT Essn FROM DEPENDENT);
 - d. SELECT * FROM EMP WHERE Salary > (SELECT MAX(Salary) FROM EMP);
-

53. In SQL, what does the predicate IS UNKNOWN apply to?

- a. Only NULL values
 - b. Boolean expressions that evaluate to UNKNOWN
 - c. Columns defined as UNKNOWN type
 - d. It is not a valid predicate
-

54. Which of the following expressions is **always** UNKNOWN?

- a. 5 = NULL
 - b. NULL IS NULL
 - c. NULL IS NOT NULL
 - d. 5 > 3
-

55. The condition $X > \text{SOME}(Q)$ is equivalent to:

- a. $X > \text{MIN}(Q)$
 - b. $X > \text{MAX}(Q)$
 - c. EXISTS (SELECT * FROM Q WHERE $X > \text{value}$)
 - d. $X \geq \text{ALL}(Q)$
-

56. Which of the following queries **cannot** be expressed without a subquery?

- a. Find employees earning more than the average
- b. Find employees with at least one dependent of the same sex

- c. Find employees in departments with more than 5 employees
 - d. All can be expressed with joins
-

57. In a query with `GROUP BY`, a correlated subquery in `SELECT` can reference:
- a. Only grouped columns
 - b. Only aggregate results
 - c. Outer query's grouped columns
 - d. Any column from outer query
-

58. The result of `NOT (UNKNOWN)` in three-valued logic is:
- a. TRUE
 - b. FALSE
 - c. UNKNOWN
 - d. NULL
-

59. Which of the following is **false**?
- a. `= ANY` is same as `IN`
 - b. `<> ALL` is same as `NOT IN` **only if no NULLs**
 - c. `> ALL (empty)` is TRUE
 - d. `EXISTS (empty)` is TRUE
-

60. To find departments with **no** employees, which is correct?
- a. `SELECT * FROM DEPT D WHERE NOT EXISTS (SELECT * FROM EMP E WHERE E.Dno = D.Dnumber);`
 - b. `SELECT * FROM DEPT WHERE Dnumber NOT IN (SELECT Dno FROM EMP);`
 - c. Both a and b — **but b fails if EMP.Dno has NULLs**
 - d. Only b
-

61. Which SQL feature is **not** discussed in Chapter 7?
- a. Triggers
 - b. Views
 - c. Indexes
 - d. Schema modification
-

62. The primary purpose of using aliases in complex queries is to:

- a. Improve performance
 - b. Reduce query length
 - c. Resolve ambiguity and enable self-joins
 - d. Bypass NULL handling
-

63. In three-valued logic, how many possible outcomes are there for `A AND B` ?

- a. 3
 - b. 6
 - c. 9
 - d. Infinite
-

64. Which of the following is **not** a valid reason for UNKNOWN results?

- a. Comparison with NULL
 - b. Empty subquery with ALL
 - c. Division by zero
 - d. Missing truth value in logic
-

65. The query:

```
SELECT * FROM R WHERE (A, B) = ANY (SELECT X, Y FROM S);
```

is:

- a. Invalid syntax
 - b. Valid and equivalent to a join
 - c. Requires S to have exactly one row
 - d. Only allowed in WHERE with EXISTS
-

66. Which operator **requires** the subquery to return a single value?

- a. IN
 - b. EXISTS
 - c. = (scalar)
 - d. ANY
-

67. If a table has 100 rows, all with NULL in column C, how many rows satisfy `C IS NULL` ?
- a. 0
 - b. 100
 - c. UNKNOWN
 - d. Error
-

68. Which of the following is **true** about `EXISTS` and NULLs?
- a. `EXISTS` returns UNKNOWN if subquery has NULLs
 - b. `EXISTS` ignores NULLs
 - c. `EXISTS` depends only on row count, not values
 - d. `EXISTS (SELECT NULL)` returns FALSE
-

69. The expression `X = ALL (empty set)` evaluates to:
- a. TRUE
 - b. FALSE
 - c. UNKNOWN
 - d. NULL
-

70. Which of the following best summarizes SQL's handling of NULL?
- a. NULL is a special value equal to zero
 - b. NULL represents missing/unknown information and uses three-valued logic
 - c. NULLs are automatically removed from tables
 - d. NULL comparisons always return FALSE
-

True/False Questions (71–100)

71. In SQL, `NULL = NULL` evaluates to TRUE.
- A. True
 - B. False
-
72. The `IS NULL` predicate can be used in the SELECT clause.
- A. True
 - B. False
-

73. A correlated subquery is executed once for the entire query.

- A. True
 - B. False
-

74. The `IN` operator is equivalent to `= ANY`.

- A. True
 - B. False
-

75. If a subquery used with `ALL` returns an empty set, any comparison with `ALL` evaluates to TRUE.

- A. True
 - B. False
-

76. Three-valued logic includes the values TRUE, FALSE, and NULL.

- A. True
 - B. False
-

77. `NOT IN` is always safe to use even when the subquery may return NULLs.

- A. True
 - B. False
-

78. Tuple comparisons in SQL require parentheses around the attribute lists.

- A. True
 - B. False
-

79. The `EXISTS` operator returns TRUE if the subquery returns at least one row, regardless of NULLs.

- A. True
 - B. False
-

80. In three-valued logic, UNKNOWN OR TRUE evaluates to UNKNOWN.

- A. True
 - B. False
-

81. Every NULL value is considered equal to every other NULL value in SQL.

- A. True
 - B. False
-

82. The condition column <> NULL is the correct way to test for non-NULL values.

- A. True
 - B. False
-

83. A scalar subquery must return exactly one column and at most one row.

- A. True
 - B. False
-

84. > ALL (empty) evaluates to FALSE.

- A. True
 - B. False
-

85. Aliases are mandatory in all SQL queries involving joins.

- A. True
 - B. False
-

86. NOT (X IN (Q)) is always equivalent to X NOT IN (Q) .

- A. True
 - B. False
-

87. In SQL, UNKNOWN and NULL are the same thing.

- A. True
- B. False

88. The query `SELECT * FROM T WHERE A = (SELECT B FROM S)` fails if S has duplicate B values.

- A. True
 - B. False
-

89. `IS NOT UNKNOWN` is a valid SQL predicate in standard SQL.

- A. True
 - B. False
-

90. Correlated subqueries can reference attributes from any level of nesting.

- A. True
 - B. False
-

91. `FALSE AND UNKNOWN` evaluates to FALSE.

- A. True
 - B. False
-

92. `NULL IS NULL` evaluates to TRUE.

- A. True
 - B. False
-

93. The `SOME` quantifier is not part of standard SQL.

- A. True
 - B. False
-

94. `= ALL` with an empty subquery returns TRUE.

- A. True
 - B. False
-

95. Three-valued logic was introduced to handle missing information consistently.

- A. True
 - B. False
-

96. `NOT EXISTS (subquery)` is unaffected by NULLs in the subquery.

- A. True
 - B. False
-

97. Tuple variables (aliases) improve query readability and prevent ambiguity.

- A. True
 - B. False
-

98. `X > ANY (Q)` means X is greater than the maximum value in Q.

- A. True
 - B. False
-

99. In SQL, an empty subquery in `IN` causes the condition to be FALSE.

- A. True
 - B. False
-

100. Chapter 7 covers aggregate functions and grouping as part of complex queries.

- A. True
 - B. False
-

Answer Key

Multiple Choice (1–70)

- 1. c
- 2. b
- 3. c
- 4. b
- 5. c

6. a

7. c

8. c

9. a

10. c

11. a

12. c

13. c

14. b

15. b

16. b

17. b

18. b

19. a

20. d

21. a

22. a

23. a

24. b

25. c

26. d

27. c

28. b

29. c

30. d

31. a

32. a

33. a

34. d

35. b

36. a

37. b

38. c

39. b

40. a

41. b

42. a

43. b

44. a

- 45. d
- 46. a
- 47. c
- 48. b
- 49. a
- 50. b
- 51. c
- 52. b
- 53. b
- 54. a
- 55. c
- 56. c
- 57. c
- 58. c
- 59. d
- 60. c
- 61. c
- 62. c
- 63. c
- 64. c
- 65. b
- 66. c
- 67. b
- 68. c
- 69. a
- 70. b

True/False (71–100)

- 71. B
- 72. A
- 73. B
- 74. A
- 75. A
- 76. B (*It's TRUE, FALSE, UNKNOWN — not NULL*)
- 77. B
- 78. A
- 79. A

- 80. B
 - 81. B
 - 82. B
 - 83. A
 - 84. B
 - 85. B
 - 86. B (*Not equivalent when NULLs present*)
 - 87. B
 - 88. A
 - 89. B (*Not standard in all SQL dialects; not emphasized in Ch7*)
 - 90. A
 - 91. A
 - 92. A
 - 93. B (*SOME is standard and equivalent to ANY*)
 - 94. A
 - 95. A
 - 96. A
 - 97. A
 - 98. B
 - 99. A
 - 100. A
-